LLM-Reasoned Motion Retrieval for Real-Time Conversational Avatars

Sardor Nodirov
Colby College

May 2026

Abstract

Conversational AI is moving from text and speech interfaces into embodied interaction. This shift changes the standard for successful interaction. A response may be linguistically fluent and still fail as embodied communication if the avatar appears delayed, emotionally mismatched, or inattentive to the user's turn. As conversational agents become more fluent in language, failures of embodied timing and attention become more visible.

This thesis introduces *LLM-reasoned motion retrieval*, an architecture for real-time conversational avatars. The system inserts a structured behavioral policy between dialogue understanding and animation. A language model reads a compact state packet describing the live interaction and emits a short-horizon behavioral directive. This directive represents the avatar's social role in the moment and guides how that role should be expressed through visible behavior. A motion layer then realizes the directive through retrieval, composition, retargeting, and browser rendering.

The contribution is an executable middle layer for embodied conversational control. It builds on the long-standing separation between communicative intent and behavior realization in embodied-agent research [7], then adapts that separation to contemporary LLM-driven avatar systems. The implementation validates the control path through typed directives, structured model calls, turn arbitration, retrieval-backed scheduling, and real-time avatar rendering. Prior MoMask and EMAGE implementations provide working motion-generation foundations for text-to-motion and co-speech behavior, grounding the proposed architecture in existing avatar systems [2, 10].

The broader claim is that real-time conversational avatars require a behavioral layer that can reason over the state of an interaction before motion is generated. By separating social intent from motion realization, the architecture gives language models a role suited to conversational reasoning while allowing motion systems to handle the physical details of embodied expression.

1 Introduction

Conversational agents are increasingly expected to act through visible bodies. A tutoring avatar, a virtual companion, a customer-support character, or an AI performer is judged through timing and presence as much as verbal correctness. Users interpret gaze, posture, facial affect, gesture scale, and silence as social evidence. These cues matter because an embodied agent appears to share the user's conversational space.

Most avatar systems still inherit a modular pipeline. A dialogue model produces text, a speech system produces audio, a co-speech model generates motion during the avatar's own utterance, and a renderer applies animation to a character. Each component can be strong in isolation, yet the combined avatar can still lack a stable account of what it is socially doing at a given moment. The

missing layer is a real-time behavioral policy that chooses an embodied role before any specific motion is selected.

The proposed architecture addresses that layer. LLM-reasoned motion retrieval uses a language model as a state reasoner over live conversational context. The reasoner receives a compact state packet summarizing the dialogue, speech dynamics, and avatar state, then emits a short-horizon behavioral directive. This directive describes the avatar's intended social behavior at a level above motor control. It may guide the avatar to keep listening, prepare a response, or acknowledge the user's turn. The motion layer translates the directive into visible animation through timing, clip selection, and facial control.

This design is motivated by the gap between social judgment and motion realization. Audio-driven co-speech systems such as EMAGE generate holistic speech-aligned behavior from audio and masked gesture context, making them strong substrates during the avatar's speaking turn [10]. Text-to-motion systems such as MDM, MotionGPT, and MoMask show that language can control human motion at the level of semantic descriptions [2, 3, 5, 14]. Part-level methods such as FrankenMotion show that body regions can be described and composed separately [8].

The need for this layer becomes clearest when the avatar is silent. Co-speech generation improves spoken output, while many of the hardest problems in conversational presence occur between speaking turns. The user may hesitate, recover from an interruption, or wait to see whether the system is still attending. These moments are brief, yet they carry real social weight. A pause after an emotional disclosure should shape the avatar differently from a pause after a factual question. The visible change may be small: a softer expression, reduced gesture amplitude, or a slightly delayed reply. Before the system chooses a motion, it has to decide what the avatar is trying to communicate.

This thesis makes that decision explicit. At runtime, the behavioral directive records the avatar's immediate social intention in a form that the rest of the system can inspect and execute. Once that intention is represented, motion generation becomes a realization problem. During speech, the system can rely on EMAGE for co-speech behavior. During listening, it can draw on part-level retrieval. Facial affect can be controlled through presets. Interruption handling remains governed by a deterministic arbiter. Each component has a separate implementation role, while the directive keeps the overall behavior aligned with a single conversational policy.

The research question follows from this division of labor: how can a real-time conversational avatar use language-model reasoning to choose socially appropriate behavior while relying on retrieval and composition for physically plausible motion? This thesis answers that question through a modular control architecture. Event-triggered reasoning selects the next behavioral directive. Part-aware retrieval finds candidate motions. Coherence-aware composition assembles them into visible behavior, and deterministic arbitration preserves turn-level control.

2 Background and Related Work

The architecture follows a long line of embodied-agent research that treats communicative behavior as a layered process. SAIBA separated intent planning, behavior planning, and behavior realization, with Function Markup Language representing communicative intent and Behavior Markup Language representing behavior to be realized by an agent [7]. That separation identifies the same boundary targeted in this thesis: the boundary between social meaning and physical expression. The setting here differs because the system is a live LLM-driven avatar stack with streaming speech, retrieval indexes, browser rendering, and interruption handling.

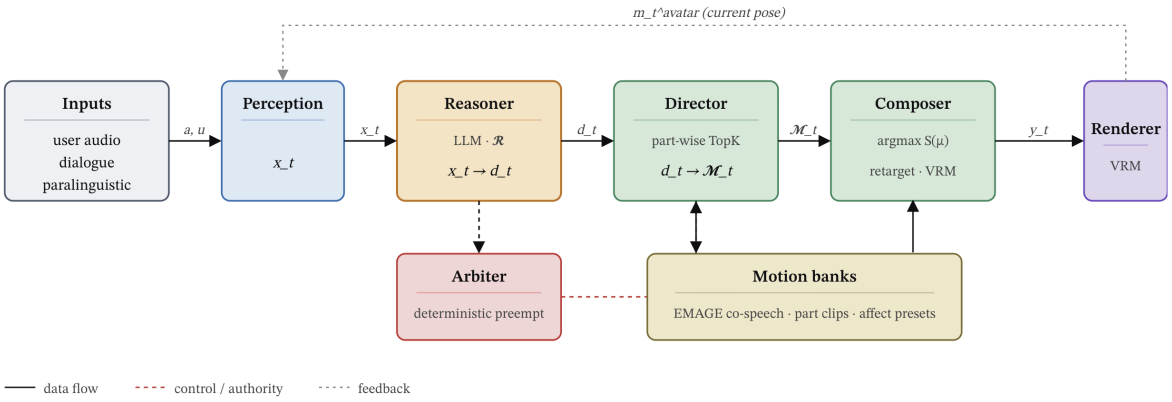


Figure 1: Proposed pipeline. The perception module compresses dialogue, audio, and avatar state into a state packet xt; a structured-output language model maps it to a behavioral directive dt; the director and composer realize dt against motion banks, with a deterministic arbiter holding control authority over the director on speech edges.

2.1 Body and Face Representation

Body and face representations shape the technical space in which such an avatar can move. SMPL-X provides a parametric model with body pose, hands, facial expression, and body shape in a unified representation [11]. FLAME provides a face model for head pose, jaw, neck, and expression, which has made it useful for facial animation and expression transfer [9]. These representations matter because conversational behavior is distributed across the body. A listener may communicate attention through head orientation, torso angle, restrained hands, and a matched facial expression at the same time.

This thesis assumes a humanoid rig whose major expressive regions can be controlled separately. The avatar may show concern through the face while keeping the arms still, or acknowledge the user with a small head movement while the torso remains in a quiet listening posture. A whole-body motion clip can contain these signals as a single package, although the package may fit the dialogue only approximately. Part-level control gives the system a more flexible way to express the intended mixture.

2.2 Audio-Driven Co-Speech Generation

Audio-driven co-speech systems build holistic motion generation on top of the mesh-based representations already introduced. EMAGE is the most direct example. Trained on BEAT2, a speech-motion corpus with body and face annotations, it produces coordinated full-body and facial motion from audio and masked gesture inputs [10]. For the present thesis, EMAGE provides a strong model of speech-driven motion during the avatar's own speaking turn, and the directive layer treats it as the default substrate while the avatar is talking.

This creates the main limitation of holistic co-speech for the present architecture. EMAGE-style output is valuable during speech because it preserves coordination across the body and face, but a conversational directive may require local adaptation. The system may need to preserve the speech-aligned torso rhythm while changing gaze, facial affect, or gesture amplitude. Part-level control provides this adaptation layer without discarding the coherence of the generated clip.

2.3 Text-to-Motion and Timeline Control

Text-to-motion research supplies the language-conditioned motion foundation. HumanML3D established a large paired resource for diverse natural human motion from text [3]. MDM showed that diffusion models can generate human motion sequences from text and other conditions [14]. MotionGPT treated human motion as a tokenized language, allowing text and motion to be modeled through a shared sequence view [5]. MoMask advanced masked modeling of 3D human motions through discrete motion tokens and transformer-based generation [2]. These systems make motion controllable through language; this thesis focuses on the prior conversational decision that selects the right motion intention.

Recent work has moved motion control closer to the needs of interactive systems. DartControl addresses real-time sequential text-driven motion through autoregressive motion primitives [17]. Multi-track timeline control lets motion be specified across temporal tracks [12]. FrankenMotion introduces part-level generation and composition, which is directly relevant to conversational avatars because the torso, arms, head, and lower body often carry different social functions in the same moment [8].

The control vocabulary of these systems is usually action-centered. A prompt says that a person walks, waves, sits, reaches, or turns. Conversation often requires a different vocabulary. The relevant description may be closer to a social stance: leave room for the user, acknowledge uncertainty, prepare to answer, or remain warmly present. LLM-reasoned motion retrieval uses language models to translate conversational context into this social vocabulary, then passes the resulting directive to a motion layer that can search, compose, and retarget physically grounded movement.

2.4 Retrieval-Augmented Motion

Subtle conversational motion leaves little margin for error. Viewers quickly notice a gesture that arrives late, a posture that feels held too long, or a transition that breaks bodily continuity. Retrieval-augmented motion methods address this problem by grounding generation in existing motion examples, which improves plausibility at the cost of some generative flexibility. ReMoDiffuse applies retrieval guidance to text-to-motion diffusion, while MoRAG and ReMoGPT extend retrieval toward part-level motion-language modeling [6, 13]. This line of work motivates the retrieval design used in this thesis: the system first gathers plausible motion candidates, then evaluates whether those candidates fit the current directive and each other.

Live conversation makes this compatibility problem more demanding than standard retrieval benchmarks. An arm movement, torso posture, and head motion may each match the directive in isolation. Rendered together, they can still produce a body that feels incoherent: the shoulder may become physically implausible, the face may conflict with the posture, or the transition may miss the conversational beat. The composer addresses this failure mode by scoring candidate combinations before the clips are accepted. Its purpose is to preserve the directive's social meaning while making the avatar appear to move as one continuous body.

2.5 Facial Affect

Facial affect provides a parallel expressive channel. MEAD is a large emotional talking-face dataset with actors recorded across emotions, intensity levels, and view angles [15]. In the proposed avatar pipeline, MEAD-style structure is useful as a source for affect presets or blendshape trajectories that can be adapted to VRM expression controls. The face layer is integrated with the directive because expression and body state need to agree for the avatar to read as socially coherent.

2.6 Language Models as Behavioral Planners

LLM-based behavioral planning connects these motion systems to conversation. Social Agent is especially close in motivation because it uses conversational LLM agents to synthesize dyadic nonverbal behavior that is both realistic and contextually appropriate [16]. The present thesis shares the emphasis on context-sensitive nonverbal behavior, while focusing on a retrieval-oriented control interface for a live browser avatar. The LLM is therefore a social policy component: it chooses the avatar's short-term behavioral commitment, while retrieval and rendering systems realize the motion.

Social Agent also illustrates a useful design tension. One path makes the language model produce increasingly detailed motor constraints. Another path asks the language model for social intent and leaves physical realization to motion systems. This thesis takes the second path. The reason is pragmatic: the LLM is strongest at interpreting conversation, while the motion system is stronger at satisfying timing, kinematics, and rig constraints. The behavioral directive is the boundary between those two strengths.

The closest older architectural ancestor is SAIBA. SAIBA separated the generation problem into communicative intent, behavior planning, and behavior realization, with Function Markup Language representing what an agent intends to communicate and Behavior Markup Language representing behavior that an animation system can realize [7]. That separation is useful here because conversational avatars still need a boundary between social meaning and physical execution. The computational setting has changed. SAIBA was designed for embodied conversational agents before current large language models, streaming speech systems, and browser-based avatar runtimes. LLM-reasoned motion retrieval preserves the same conceptual boundary while changing the policy mechanism and the actuator stack.

This relationship clarifies why the directive is central. A prompt sent directly to a motion model can describe a gesture, posture, or action. A directive describes the social role of the avatar at a particular moment in a conversation. The object contains timing, priority, gaze, affect, body-region guidance, and vocalization authority in one validated form. It is therefore closer to a modern behavior-plan object than to a text-to-motion caption. That distinction matters during interruption and silence, where the avatar's primary task is to manage the conversation floor.

U-Mind and A^2-LLM point toward integrated audiovisual agents that respond in real time [1, 4]. The present thesis takes a systems route through this space. It uses LLM reasoning where contextual judgment is needed, then asks retrieval, composition, retargeting, and rendering to handle embodied realization. This route gives the project a practical inspection surface during development and evaluation.

3 Research Gap

Real-time conversational avatars face constraints that isolated motion-generation tasks rarely capture. In a live exchange, the avatar must react to speech boundaries as they happen: when the user begins speaking, stops speaking, interrupts, hesitates, or remains silent. These moments require different visible behavior. Sometimes the avatar should stay still because movement would feel intrusive. At other times, stillness itself can make the system appear absent.

Existing work provides strong pieces of this problem. Co-speech models generate motion aligned with the avatar's own speech. Text-to-motion models make movement controllable through language. Part-level methods increase control over specific body regions, and facial datasets provide a basis for affective expression. Recent LLM-based systems show that language models can interpret dialogue context and guide embodied behavior. What remains underdeveloped is the interface between live

conversational state and motion generation: a representation that says what the avatar is socially trying to do before the system decides how that behavior should move.

That interface has to satisfy several requirements at once. It must be fast enough to handle speech edges, structured enough to validate, expressive enough to represent social intent, and concrete enough to support retrieval. It must also coexist with deterministic control, since conversational events may arrive while a model call is still running. This thesis proposes the behavioral directive as that interface.

Three design pressures follow from this gap. The first is timing. A language-model call may be suitable for deciding how to respond after a pause, but the first visible reaction to an interruption often has to happen immediately. The system therefore needs both deliberative reasoning and low-latency reflexes. The second pressure is continuity. A user experiences speaking, listening, thinking, and silence as one exchange, so the avatar's visible behavior should also be managed as a continuous stream. The third pressure is inspectability. When the avatar behaves poorly, the system should make it possible to locate the failure: the directive may have represented the wrong social intent, retrieval may have returned weak candidates, composition may have accepted an incompatible motion, or rendering may have introduced an artifact.

The research contribution is organized around these pressures. The directive provides an inspectable representation of social intent. Retrieval grounds visible behavior in motion data. Coherence scoring protects physical continuity across separately selected motion parts. The arbiter handles moments where conversational timing changes faster than the reasoner can complete. Together, these components define a control architecture for real-time embodied conversation.

4 Methods

The method proposed here is an LLM-reasoned motion retrieval architecture for live conversational avatars. Its central contribution is a real-time control pipeline that inserts an interpretable social-behavior policy between dialogue understanding and motor realization. Prior frameworks such as SAIBA are useful historical context because they separated communicative function from behavior realization [7]. The contribution is the executable modern form of that separation: continuous event reasoning, a directive schema designed for retrieval, heterogeneous motion banks, shared-joint composition scoring, and turn arbitration across speaking, listening, and silence.

4.1 System Overview

The system assumes a base conversational stack with automatic speech recognition, an LLM dialogue engine, text-to-speech, an audio-driven co-speech backend, and a browser-rendered VRM character. At wall-clock time t, the proposed layer maps the current interaction into a compact state packet, produces a behavioral directive, converts that directive into motion candidates, and renders a composed motion plan. The architecture is summarized as

$$\begin{aligned} x_t &= \mathcal{P}(u_{1:t}, a_{1:t}, m_t^{\text{avatar}}), \\ d_t &= \mathcal{R}(x_t), \\ \mathcal{M}_t &= \mathcal{D}(d_t), \\ y_t &= \mathcal{C}(\mathcal{M}_t, m_t^{\text{avatar}}, e_t^{\text{speech}}, s_t^{\mathcal{A}}), \end{aligned} \tag{1}$$

where $u_{1:t}$ is dialogue history, $a_{1:t}$ is the recent audio and event stream, m_t^{avatar} is the current rendered motion state, e_t^{speech} is the audio-driven co-speech stream when the avatar is speaking, and y_t is the final pose, blendshape, gaze, and schedule sent to the renderer.

4.2 Perception State Packet

The perception normalizer sits between high-frequency interaction events and the state reasoner. Its role is to convert continuous streams of text, audio, timing, and avatar-control events into a compact representation that a language model can reason over at conversational timescales. Each reasoning call receives a state packet

$$x_t = \langle H_t,\ q_t,\ \nu_t,\ \sigma_t,\ \eta_t,\ r_t,\ m_t,\ \ell_t \rangle,$$

where H_t stores bounded dialogue history and q_t stores the latest utterance. The remaining fields describe the live interaction state: ν_t records voice activity, σ_t records silence timing, η_t summarizes paralinguistic evidence, r_t stores the trigger for the reasoning call, m_t describes the avatar's active motion, and ℓ_t records latency and in-flight request status. The paralinguistic field is symbolic rather than audio-continuous. It can encode affective direction, arousal, confidence, pitch or energy shifts, filled pauses, and interruption evidence as compact cues for social reasoning. These cues help the reasoner infer the avatar's conversational state while leaving motor control to the realization layer.

This compression is necessary because the relevant signals arrive at different rates. Voice activity may change within a few frames, silence accumulates over hundreds of milliseconds, dialogue history changes by utterance, and avatar motion changes continuously during rendering. Passing those streams directly into a language model would create unstable inputs and unnecessary latency. The normalizer instead converts them into comparable state fields. Voice activity becomes a floor-ownership signal. Silence duration becomes evidence about conversational pressure. Recent text provides the local discourse context. Paralinguistic features become a short affective summary in place of a full audio trace. Active motion records what the avatar is already doing, which helps the reasoner avoid directives that would clash with the visible pose.

The same packet also serves as the system's logging unit. Every behavioral directive can be traced back to the state summary that produced it. This is useful during evaluation because a poor avatar response can originate at several points in the pipeline. If x_t correctly captured the user's utterance, timing, affective evidence, and current avatar state, the error likely occurred in reasoning, retrieval, composition, or rendering. If x_t missed a speech edge or flattened an important affective cue, the failure belongs to perception. The state packet therefore supports both runtime control and post-hoc diagnosis.

4.3 Behavioral Directive Schema

The state reasoner maps each normalized state packet to a behavioral directive,

$$d_t = \langle s_t,\ i_t,\ v_t,\ b_t,\ f_t,\ g_t,\ h_t,\ \pi_t,\ \chi_t \rangle.$$

The directive gives the rest of the system a compact description of what the avatar should do next. The state field s_t identifies the current conversational regime, while the intent field i_t records the social function of the behavior. When the avatar should speak, v_t carries the response text. The body directive b_t specifies whole-body or part-specific guidance, and the face directive f_t describes affective expression and gaze. The gesture field g_t supports phrase-anchored beats during speech. The horizon h_t defines how long the directive remains valid, π_t controls preemption, and χ_t stores trace information for debugging and later analysis.

This directive is designed as a retrieval-oriented control object. It gives the motion layer more structure than an ordinary natural-language prompt while staying above low-level motor control. For example, a listening directive may describe restrained supportive attention, with a slight forward

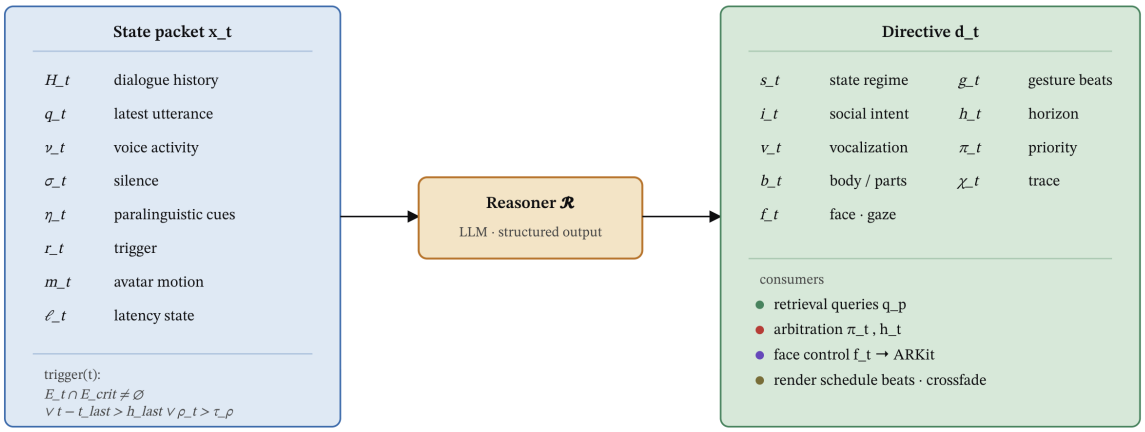


Figure 2: Mapping from state packet to behavioral directive. The reasoner consumes the validated packet xt and emits a structured short-horizon directive dt whose fields are routed to retrieval, arbitration, face control, and rendering.

torso bias, relaxed arms, a small acknowledgment nod, warm facial concern, and soft eye contact. Those fields become the query and scheduling surface for motion retrieval. The motion layer can then search for candidate clips, adapt the relevant channels, and schedule the result within the current turn state.

The schema also keeps social reasoning readable. A directive can specify that the avatar should listen with restrained concern, hold a stable torso, keep the arms quiet, orient toward the user, and use a low-intensity sad or neutral expression. It leaves BVH file selection, quaternion values, blendshape curves, and transition details to the realization layer. This separation makes the directive understandable to a human evaluator while still giving the runtime enough information to retrieve, compose, and schedule motion.

The contract defines enumerated agent states, emotion labels, intensity levels, gaze targets, priorities, trigger kinds, and nested body and face directives. The horizon is clamped to a short range, so each directive remains tied to the current conversational moment. The trace field is kept compact, which makes it useful for debugging without bloating the runtime object. In practice, the schema functions as both a live-system artifact and a conceptual account of the architecture's control layer.

4.4 State Reasoner

The state reasoner $\mathcal{R}$ is the policy component of the system. Its purpose is to select the next short-lived social-behavior commitment. It is constrained by the directive schema, which makes several classes of failure explicit before the motion layer receives them. A directive with an invalid state, missing horizon, unsupported affect label, or illegal priority value can be rejected immediately. This is important because downstream motion systems need stable contracts in a validated format.

The reasoner consumes bounded context. Recent dialogue is preserved, while older turns are summarized into motion-relevant facts such as unresolved user affect, task state, or conversational commitments. The reasoner also receives the current avatar state and the last directive. This prevents locally sensible decisions that are unstable in motion. For example, the model may want

to schedule an expressive arm gesture, but the current arm layer may already be fading from a prior response. The state packet gives the model enough information to avoid issuing that conflicting command.

The intended reasoning problem is deliberately narrow. The reasoner selects whether the avatar should listen, backchannel, think, speak, remain idle, or gently re-enter. It also decides which expressive channels should carry that choice. In emotionally delicate moments, stillness and low amplitude may be the correct behavior. In a factual explanation, a small semantic beat may help. In a hesitation, the avatar may need to hold attention and delay speech. The schema forces these judgments into a form that can be logged and evaluated.

During development, the reasoner can use a frontier model. For deployment, the same logged triples (x_t, d_t, y_t) can train a smaller behavioral policy. This is one of the practical advantages of a directive layer. The output becomes an explicit training target with a fixed schema.

4.5 State Reasoning Cadence

The state reasoner is invoked by conversational events. Speech edges, TTS edges, horizon expiry, gesture completion, salience changes, and idle heartbeat events can all request a new directive. Let E_t be the set of events observed since the last reasoning call. A new call is scheduled when

$$\text{trigger}(t) = \mathbb{1}\left[E_t \cap E_{\text{critical}} \neq \emptyset \ \vee \ t - t_{\text{last}} > h_{t_{\text{last}}} \ \vee \ \rho_t > \tau_\rho\right], \tag{2}$$

where ρ_t is a salience score computed from paralinguistic changes and τ_ρ is an adaptive threshold. Critical events include user barge-in, TTS completion, explicit action requests, and directive expiry.

4.6 Conversational State Regimes

The state set is small by design. These states function as authority regimes for the motion system.

In the speaking state, the avatar is producing speech. Audio-driven co-speech has high authority over the arms and torso because speech rhythm provides a strong timing signal. The directive still influences posture bias, facial affect, gaze, and sparse semantic beats. This lets EMAGE or a similar substrate carry the motion rhythm while the behavioral policy shapes the social reading of the response.

In attentive listening, the user is speaking and the avatar should display stable attention. The motion is restrained. Gaze and posture matter more than large gestures. The avatar follows the conversational floor and shows that it remains present and available.

Backchannel listening is more active. The avatar may nod, briefly change expression, or produce a short acknowledgment. The important constraint is turn ownership. A backchannel should support the user's turn and preserve floor ownership, so this state depends heavily on the arbiter.

Thinking covers the interval after a user stops and before the avatar is ready to answer. A slight gaze-away, lower gesture amplitude, or restrained posture shift can keep the avatar from appearing frozen. Proactive engagement covers a longer silence in which the avatar may begin a gentle prompt or visibly prepare to re-enter. Idle neutral is the low-pressure state in which the avatar preserves presence while the interaction is quiet.

4.7 Motion Banks and Offline Indexing

The motion director retrieves from specialized sources because conversational embodiment spans different behavioral regimes. EMAGE-like co-speech generation supplies audio-conditioned behavior during speech [10]. Part-level body clips provide listener, thinking, silence, and posture behavior,

following the motivation of part-level motion composition [8]. MEAD-style data supplies affect references for expression presets [15]. The body is decomposed into canonical regions

$$\mathcal{P} = \{\text{TORSO}, \text{ARMS}, \text{HEAD}, \text{LOWER}\}. \tag{3}$$

Offline preprocessing converts motion corpora into retrieval-ready part clips. Each source clip is represented as

$$\mu = \langle \Theta_{1:T},\ \phi,\ \psi,\ \delta,\ \omega,\ \zeta \rangle, \tag{4}$$

where $\Theta_{1:T}$ is the motion trajectory, ϕ is the full-clip caption, ψ contains part captions, δ stores duration and frame-rate metadata, ω stores boundary-joint statistics, and ζ stores source, license, representation, and retargeting metadata. A part clip is the projection of μ onto a region p:

$$\mu^p = \langle \Theta^p_{1:T},\ \psi^p,\ \mathbf{e}(\psi^p),\ \omega^p,\ \Theta^p_1,\ \Theta^p_T \rangle, \tag{5}$$

where $\mathbf{e}(\cdot)$ is a text embedding function. Separate vector spaces are stored for whole-body captions and for each body region. This preserves semantic modularity while retaining kinematic information for later compatibility scoring.

In the local implementation, this preprocessing appears in the factor index and HumanML3D asset materialization pipeline. The system stores the annotation order, embedding status, SQLite retrieval index, and HumanML3D motion files with their associated metadata. These artifacts turn research datasets into runtime assets: motions can be embedded once, indexed once, and queried repeatedly during interaction. This way, the system needs short retrieval calls rather than offline batch processing.

HumanML3D uses a compact, structured motion representation. It includes root motion, relative joint positions, continuous rotations, local velocities, and foot-contact flags in a fixed layout. The composition utility uses that structure directly. It maps HumanML3D joints into body regions such as torso, arms, head, and legs, then trims selected clips to a common frame count before combining them. For each region, it copies the corresponding joint channels from the selected source clip. When lower-body motion is available, it also preserves root motion and foot-contact ownership from that source, which helps maintain balance and avoid inconsistent foot behavior.

4.8 Online Retrieval and Coherence-Aware Selection

Given directive d_t, the director creates query strings for each body region. If an explicit part field exists, that field is used. Otherwise the query falls back to the whole-body directive plus the intent label:

$$q^p_t = \begin{cases} b^p_t & \text{if } b^p_t \neq \emptyset, \\ b^{\text{whole}}_t \oplus i_t & \text{otherwise.} \end{cases} \tag{6}$$

The director retrieves a candidate set from each part index:

$$\mathcal{C}_p = \text{TopK}_{\mu^p \in \mathcal{I}_p} \cos\left(\mathbf{e}(q^p_t), \mathbf{e}(\psi^p)\right). \tag{7}$$

Retrieval is candidate generation. The composer selects one candidate per part by optimizing

$$\mu^* = \arg \max_{\mu \in \prod_{p \in \mathcal{P}} \mathcal{C}_p} S(\mu, d_t, m_t), \tag{8}$$

where

$$
\begin{aligned}
S(\mu, d_t, m_t) = \lambda_{\text{rel}} \sum_{p \in \mathcal{P}} \cos\Big(\mathbf{e}(q_t^p), \mathbf{e}(\psi_{\mu_p}^p)\Big) \\
+ \lambda_{\text{glob}} \cos\Big(\mathbf{e}(b_t^{\text{whole}}), \mathbf{e}(\Phi(\mu))\Big) \\
- \lambda_{\text{bdry}} \sum_{(p,q) \in \mathcal{B}} C_{\text{bdry}}(\mu_p, \mu_q) \\
- \lambda_{\text{pose}} \sum_{p \in \mathcal{P}} C_{\text{pose}}(\mu_p, m_t) \\
- \lambda_{\text{style}} C_{\text{style}}(\mu).
\end{aligned} \tag{9}
$$

Here $\Phi(\mu)$ is a summary of selected part captions and $\mathcal{B}$ is the set of adjacent body-part boundaries. The boundary penalty is

$$
C_{\text{bdry}}(\mu_p, \mu_q) = \frac{1}{|J_{p,q}|} \sum_{j \in J_{p,q}} \left[\left\| \dot{\theta}_p^j(0) - \dot{\theta}_q^j(0) \right\|_2^2 + \kappa \left\| \theta_p^j(0) \ominus \theta_q^j(0) \right\|_2^2 \right], \tag{10}
$$

where $J_{p,q}$ contains shared or adjacent joints, θ^j is joint rotation, $\dot{\theta}^j$ is angular velocity, and $\ominus$ denotes quaternion log-distance. The pose-transition cost compares the candidate start pose with the current rendered pose:

$$
C_{\text{pose}}(\mu_p, m_t) = \frac{1}{|J_p|} \sum_{j \in J_p} \left\| \theta_{\mu_p}^j(0) \ominus \theta_{m_t}^j \right\|_2^2. \tag{11}
$$

The style cost penalizes combinations whose energy, amplitude, or posture statistics diverge:

$$
C_{\text{style}}(\mu) = \text{Var}_{p \in \mathcal{P}}(E_{\mu_p}) + \text{Var}_{p \in \mathcal{P}}(A_{\mu_p}) + \text{Var}_{p \in \mathcal{P}}(P_{\mu_p}). \tag{12}
$$

The retrieval query code follows the same structure. It embeds the directive fields, retrieves top candidates for each part, evaluates torso-arm-head combinations, and adds a deterministic seeded tie break so user-study recordings can be reproduced. The scoring metadata is preserved, which makes it possible to inspect whether a chosen motion won because it was semantically strong, physically smooth, or the best available candidate under a sparse index.

The current implementation evaluates torso, arms, and head in the online query path, with legs handled in the composition utility and scheduling logic. This reflects the practical priority of humanoid characters: upper-body behavior carries most visible turn-taking and affect, while lower-body channels stabilize root motion and contact behavior.

4.9 Layered Composition and Speaking-State Realization

After selection, the composer converts part clips into a layered schedule. Each body part has an owner at time t:

$$
o_t^p \in \{\text{RETRIEVAL}, \text{EMAGE}, \text{IDLE}, \text{GESTURE}, \text{HOLD}\}. \tag{13}
$$

During avatar speech, EMAGE owns the rhythmic torso and arm layer unless a high-priority semantic beat temporarily overlays part of the motion. During listening, retrieval owns the head, torso, face, and usually the arms. The final pose for joint j is computed as a normalized blend:

$$
\theta_t^j = \text{Normalize}\left(\bigoplus_{r \in \mathcal{R}_t^j} w_{r,t}^j \theta_{r,t}^j \right), \tag{14}
$$

where $\mathcal{R}_t^j$ is the active source set, $w_{r,t}^j$ is the source weight, and $\oplus$ denotes quaternion blending. Temporal transitions use a smooth crossfade,

$$w_{\text{new}}(\tau) = 3\tau^2 - 2\tau^3, \quad \tau = \text{clip}\left(\frac{t - t_0}{w}, 0, 1\right). \tag{15}$$

When $s_t = \text{SPEAKING}$, EMAGE or an equivalent substrate produces the base pose stream

$$e_t^{\text{speech}} = \mathcal{E}(\text{audio}_{t:t+\Delta}). \tag{16}$$

A phrase-anchored gesture beat (c_j, β_j) is mapped to wall-clock time through TTS alignment, then inserted as

$$y_t^{\text{arms}} = (1 - \alpha_j(t))\, e_t^{\text{arms}} + \alpha_j(t)\, \mu_{\beta_j}^{\text{arms}}(t - t_j), \tag{17}$$

where $\alpha_j(t)$ is active only near the beat window. This keeps speech rhythm under audio-driven control while allowing retrieval to express discourse-specific meaning.

4.10 Listening and Silence Realization

Listening and silence are the regimes where the architecture matters most. In these states outgoing speech audio is absent, so a co-speech model has weak conditioning for motion. A purely audio-driven avatar may freeze, fall back to a generic idle loop, or continue a movement pattern that mismatches the interaction. The directive layer gives the system a way to decide what kind of presence the avatar should maintain while the user has the floor.

During attentive listening, the reasoner selects behavior that indicates presence while leaving conversational space open. The retrieval layer favors low-amplitude torso and arm motion, controlled head behavior, and face or gaze settings that match the user's affect. During backchannel listening, the avatar may briefly acknowledge the user through a small nod, expression change, or short vocalization. The arbiter determines whether that acknowledgment can occur safely inside the user's turn.

Thinking is different from listening. The user's turn may have ended, yet the avatar should look active while preparing a response. A gaze shift, reduced gesture amplitude, or slight posture change can communicate processing. Proactive engagement is different again. After a longer silence, the avatar may begin to re-enter the conversation with a visible shift before speech starts. This body-first re-entry is important because sudden speech from a static pose reads as mechanical.

4.11 Turn Arbitration and Rendering

The turn arbiter $\mathcal{A}$ is deterministic. It tracks user voice activity, avatar speech state, active clips, pending directives, and priority. Let $V_t \in \{0, 1\}$ indicate user speech and let $T_t \in \{0, 1\}$ indicate avatar TTS. The arbiter state is

$$s_t^{\mathcal{A}} = \begin{cases} \text{LISTENING-ATTENTIVE} & \text{if } V_t = 1 \text{ and } T_t = 1, \\ \text{SPEAKING} & \text{if } V_t = 0 \text{ and } T_t = 1, \\ s_t & \text{otherwise.} \end{cases} \tag{18}$$

The first case handles barge-in. The arbiter can reduce speech authority, fade active gestures, and shift visible behavior toward listening before a new LLM call returns. It also applies dwell-time constraints,

$$\Delta(s) \geq \Delta_{\min}(s), \tag{19}$$

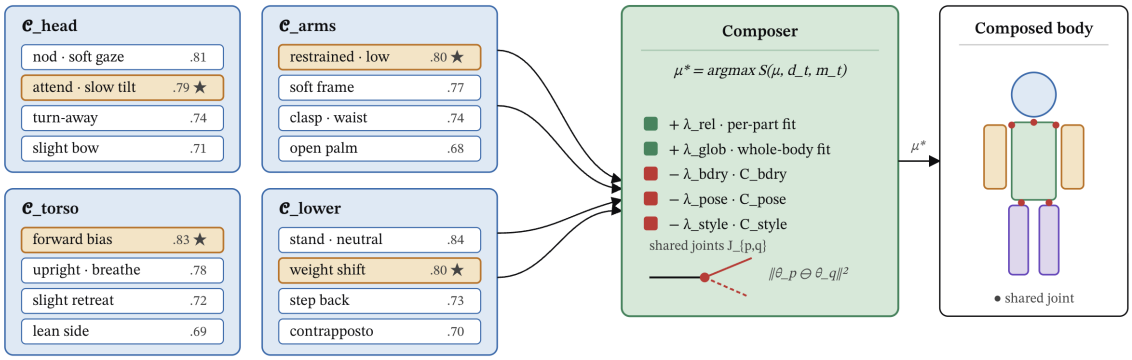


Figure 3: Part-level retrieval and coherence-aware composition. Per-region Top-K candidates (highlighted: chosen) feed the composer, which selects one candidate per part by maximizing S(, dt, mt): per-part and whole-body fit minus boundary, pose-transition, and style-variance costs evaluated at shared joints.

except during critical preemption. This prevents rapid flicker between thinking, backchannel listening, and idle behavior during transient ASR or VAD uncertainty.

Priority is handled at the directive level. Low-pressure idle behavior can yield to almost any meaningful event. High-priority interruption handling can preempt active motion, although the transition still needs a configured fade or safe cut point. This keeps the avatar responsive while avoiding visible state changes for every small signal.

The renderer receives bone rotations, blendshape weights, gaze targets, and scheduled clip events. The target character is a browser-rendered VRM humanoid using Three.js. Motion is converted from SMPL-X-style representations to VRM through a fixed bone correspondence for the humanoid body, while facial controls are mapped into the avatar's expression channels. The directive remains upstream from retargeting, so a change of avatar rig affects the realization layer while preserving the social policy.

The renderer operates on buffered motion windows. This allows clip-level blending, scheduled gesture beats, and interruption recovery. The server sends compact pose buffers and event metadata; the frontend applies local interpolation and expression smoothing so network jitter remains separate from visible motion jitter.

5 Prototype and Implementation

This section describes the current prototype and clarifies which parts of the architecture have been exercised in execution. The goal is to distinguish the validated control path from the actuator components that still require larger-scale empirical testing.

The prototype shows that live conversational state can be channelled into a stable behavioral contract without sacrificing the timing properties of an exchange. State packets enter a typed reasoning layer, leave it as validated directives, and reach a browser avatar through a single dispatch channel. The same control path that Section 1 formalizes is exercised continuously during interaction: a directive that passes schema validation arrives at the motion director carrying the fields the

reasoner committed to, so downstream modules can be developed against one contract rather than against unstructured model output.

Three observations follow from running this loop. First, structured output is essential rather than ornamental. When the reasoner emits free-form text, downstream scheduling becomes brittle and most of the engineering effort drifts into parsing and recovery; a validated directive removes that work and exposes the reasoner's behavior in a form a human can read. Second, conversational events arrive faster than reasoning calls complete, which means the arbiter must own the immediate response to speech edges while the directive layer governs slower contextual behavior. Third, the body-part decomposition is meaningful at runtime: authority over arms, torso, head, and face can be reallocated across speaking, listening, and silence without changing the directive schema, which is the operational form of the regime separation argued for in Section 4.

5.1 The Control Path

The control path is the part of the system that the prototype exercises most fully. A typed directive contract sits at its center, with enumerated states, intent labels, body and face fields, horizon, and priority. Every module that produces or consumes a directive does so through this schema. The reasoner uses provider-specific structured-output adapters so the same contract can be backed by different language models without disturbing the rest of the pipeline. This separation has practical value beyond engineering hygiene: a reasoner can be swapped, distilled, or constrained without rewriting the motion layer that depends on it.

The arbiter operates beneath the reasoner on a deterministic schedule. It tracks voice-activity edges, TTS edges, in-flight reasoning calls, and dwell-time constraints. When the user begins to speak, the arbiter does not wait for the reasoner to commit to a new directive: it reduces ongoing motion authority and shifts visible behavior toward attentive listening immediately, and the reasoner refines that response when its call returns. This division is the operational form of the timing argument from the research gap. Deliberative reasoning sits behind a low-latency reflex layer rather than blocking it.

The motion director maps directive fields onto concrete avatar resources. Head fields drive the neck, head, jaw, and eye channels; arm fields drive shoulder, arm, hand, and finger chains; torso and lower-body fields drive the corresponding bone groups; face fields drive expression presets. During speech the rhythmic torso and arm channels defer to the co-speech substrate while retrieval still governs head behavior and face control; during listening and silence the director takes wider authority because the speech substrate has weak conditioning. The runtime behavior of the system therefore changes regime by regime in the way the methods section predicts, and that change is observable in the dispatch trace rather than only asserted in the architecture diagram.

5.2 Motion Realization

Retrieval and composition are implemented far enough to exercise the architecture and to surface its remaining requirements. Offline preprocessing converts HumanML3D-style motion records and FrankenMotion-style annotations into part-indexed entries with caption embeddings, boundary statistics, and source metadata. Online queries embed the directive's body fields, retrieve per-region candidates, and score combinations against the objective in Equation 9. The individual scoring terms are preserved on each retrieved combination, which makes it possible to inspect whether a chosen motion was selected because it was semantically strong, kinematically smooth across part boundaries, or simply the best available candidate under a sparse index. That kind of diagnostic visibility is the practical payoff of writing retrieval as scored composition rather than as a single

nearest-neighbour lookup.

Composition runs directly in the HumanML3D representation. Selected part clips are trimmed to a common frame count, joints are assigned to canonical regions, and the corresponding channels are copied into a combined trajectory. Root motion and foot-contact ownership are taken from the lower-body source so that the assembled motion does not introduce inconsistent contact behavior. Deterministic tie-breaking keeps user-study trials reproducible across runs, which matters because the evaluation in Section 6 depends on identical stimuli across conditions.

The browser mixer is the renderer-facing counterpart. It schedules clips by bone mask, applies quaternion crossfades between sources, fades active motion during interruption, and routes face presets into VRM expression weights. Body-part ownership is exposed at runtime so it can be verified on the rendered avatar rather than only inferred from logs. This layer matters because a retrieval result only becomes conversational behavior after it survives scheduling, retargeting, and animation playback under live timing. A scoring objective that looks correct in isolation can still produce visibly poor motion if the mixer transitions discard the directive's intended timing, which is why the mixer is part of the validated path rather than treated as a downstream rendering detail.

5.3 Prior Substrates

Two earlier implementations supply working motion-generation evidence that the new architecture is designed to stand on. The MoMask path provides text-to-motion inference, joint recovery, and a streaming generator coordinated with LLM-selected speech, TTS, and audio-motion alignment [2]. It established the text-to-motion side of the project before the directive layer existed, and it remains usable as a semantic motion source behind the new contract. The EMAGE path provides audio-driven holistic co-speech behavior with FLAME-to-ARKit expression conversion and SMPL-X-to-VRM retargeting, delivered through the same browser avatar used by the present system [9–11]. Together these substrates show that the speaking-state and text-to-motion components the directive layer treats as resources are themselves runnable systems, so the present architecture is composing live components rather than depending on external work yet to be done.

5.4 Scope of Validation

The prototype validates the interface and the control surface around it. The behavioral contract, the structured reasoning backends, the arbiter, the indexed retrieval, the part composition utility, the face-preset dispatch, the dispatch channel, and the browser mixer exist and operate together. What remains is the scaled-up actuator: larger conversation-relevant motion banks, production-quality retargeting of retrieved clips into the live avatar, calibrated face adaptation across characters, full integration of EMAGE behind the directive layer, and the user-study evaluation laid out in the next section.

6 Evaluation Design

The evaluation tests whether directive-level reasoning improves social fit during live interaction. The primary perceptual measure is appropriateness-in-context: whether the avatar's visible behavior reads as the right thing to be doing at this moment in this conversation. Responsiveness and naturalness are secondary measures, since a motion can look plausible in isolation while fitting the conversation poorly.

The preferred design is within-subjects. Each participant sees the same scenario rendered under four system variants, with order counterbalanced. The *No-Reasoning* baseline uses a heuristic state

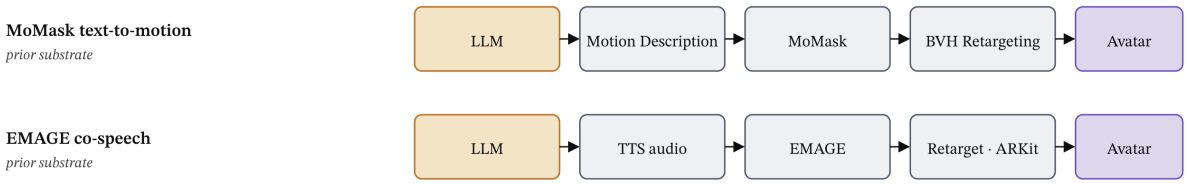


Figure 4: Prior runnable motion-generation substrates that the proposed architecture composes behind its directive contract: MoMask provides text-to-motion behind the avatar, and EMAGE provides audio-driven co-speech behavior during speaking turns.

machine with speech-driven movement during the avatar's utterances and simple idle behavior elsewhere. The *Face-Only Reasoning* variant preserves directive reasoning for face and head while restricting body retrieval, which isolates the incremental value of full-body behavior. The *Rule-Routed Retrieval* variant draws on the same retrieval sources, mapping coarse dialogue states to clips through fixed rules in place of LLM reasoning. The *Full Directive Pipeline* combines directive reasoning, part-level retrieval, coherence-aware composition, and turn arbitration.

The scenarios should be short, controlled, and socially diagnostic. A user may hesitate after a factual question, interrupt during an explanation, disclose sadness, remain silent after a light exchange, or abruptly shift topics. Each scenario should be rendered under each condition with the same user-side audio. The scenarios should last long enough for the avatar to pass through more than one regime, because the central claim concerns transitions across speaking, listening, thinking, and silence.

The hesitation scenario should test whether the avatar can preserve presence during uncertainty. The user asks a question, starts a partial follow-up, then pauses. A weak system will either freeze or begin speaking too early. The proposed system should maintain attentive posture, delay vocalization, and possibly show reflective processing after the user's turn ends. The interruption scenario tests the fastest path through the architecture. The user begins speaking over the avatar. The arbiter should fade outgoing behavior and shift to listening before a full reasoner call returns.

The emotional-disclosure scenario should test affect agreement across body and face. A supportive response should be restrained, with motion amplitude and expression intensity calibrated to the moment. The extended-silence scenario tests proactive engagement. The avatar should signal continued availability during the pause and re-enter gently after the silence threshold. The topic-shift scenario tests continuity: the avatar must release the posture and affect of the previous exchange once the user introduces a new direction.

Three hypotheses follow from this design. H1 is that the Full Directive Pipeline raises appropriateness ratings over the No-Reasoning baseline across every scenario. H2 is that the gap between conditions grows in listening and silence scenarios relative to scenarios dominated by avatar speech, because the directive layer takes wider authority over the body when the co-speech substrate has weak conditioning. H3 is that faster arbiter response time correlates with higher appropriateness ratings in the interruption scenario, since the deterministic preemption path is what the architecture asks to do most of the work there.

Perceptual measures should ask participants about responsiveness, social presence, naturalness, turn-taking quality, and appropriateness of behavior for the moment. Naturalness alone is too weak because a motion can look plausible in isolation while fitting the conversation poorly. Appropriateness is the more important construct for this thesis because the avatar's body is being judged as part of an interaction. A small nod at the right moment may be more successful than a visually complex

gesture with poor timing.

Participants can compare paired videos or live interactions. A forced-choice question should ask which avatar feels more like a natural conversational partner, with an uncertainty option. Likert ratings can capture naturalness, attention, timing, and awkwardness. Open-ended comments should also be collected because they often identify the concrete failure: late gaze, overactive hands, frozen silence, mismatched face, or abrupt transition. The pilot would enroll 20 participants, enough to expose large perceptual differences and recurring failure modes. The analysis uses paired comparisons across conditions within each scenario, with appropriateness rating as the primary dependent measure and a paired Wilcoxon test as the inferential procedure.

The heuristic condition should reveal the value of contextual state reasoning. The head-and-face condition should reveal how much full-body behavior contributes once intent is already available. The retrieval-only condition should reveal whether motion search alone is enough for conversational fit. The full condition should reveal whether reasoning, retrieval, composition, and arbitration reinforce one another. This framing keeps the study tied to the architecture and prevents a single overall rating from carrying the entire result.

7 Discussion

The strongest conceptual contribution of this architecture is the treatment of listening and silence as first-class control regimes. Many avatars look most alive while speaking because speech gives motion systems a dense conditioning signal. Real conversation also includes waiting, attending, hesitating, yielding, and preparing, and these intervals carry much of what makes an avatar feel present. By giving them explicit states, retrieval pathways, and dedicated authority over the body, the architecture treats the whole exchange as continuous embodied behavior rather than as a speaking performance bracketed by idle loops.

The modular design also creates a useful relationship with emerging end-to-end systems. U-Mind and A^2-LLM point toward unified multimodal models that may eventually handle language, speech, motion, and face inside one loop [1, 4]. The directive layer is designed to survive that transition. Even when a single model produces coordinated speech and motion, a system designer still needs an inspectable representation of what the avatar is socially committing to in order to evaluate it, debug it, and replace components beneath it. The directive provides that representation as a stable evaluation and control interface, which is a different role from the one it plays today as a glue layer between separate motion backends.

The architecture also predicts where it will fail. Directive granularity is coarse, so the system cannot express culturally specific or individually idiosyncratic listener behavior without per-user policy adaptation. When paralinguistic perception is noisy, the reasoner will commit to a directive with confident social intent built on weak evidence, and the visible failure will look like a reasoning failure even though its source is upstream. Retrieval has a quieter failure mode in sparse regions of the index, where coherence scoring may accept a combination that is internally plausible yet socially generic, and the trace will register the selection as healthy. These limitations are real, and the directive layer makes them locatable rather than masking them inside an end-to-end output.

8 Scope and Limitations

The current prototype validates the control path and part of the retrieval actuator. Full deployment requires larger motion banks, stronger retargeting, expression calibration for each avatar, latency

optimization, and user-study validation. Since the central contribution is the behavioral interface and runtime architecture, the actuator can continue to mature behind the same directive contract.

Data coverage remains a practical limitation because conversational behavior often requires subtle low-amplitude motion, while many motion datasets emphasize larger actions. A walking, dancing, exercising, or gesturing dataset may provide useful kinematics while still underserving listener posture, restrained affect, and silence behavior. The retrieval layer therefore needs either careful filtering of existing corpora or new annotation that marks conversational functions, body-part roles, and timing boundaries.

Part-level retrieval adds another data requirement. A candidate head motion, arm motion, and torso posture must agree near shared joints. FrankenMotion provides a useful direction by treating body parts as controllable units [8], yet live avatars require compatibility metadata that reaches beyond semantic captions. The index needs boundary pose, velocity, style, amplitude, and contact information so the composer can reject combinations that are semantically appealing yet physically awkward.

Retargeting is also difficult. SMPL-X, HumanML3D, BVH, FLAME, ARKit blendshapes, and VRM bones encode bodies and faces in different spaces [9, 11]. The local EMAGE work provides a concrete SMPL-X to VRM path and FLAME to ARKit conversion, and retrieved body clips still need careful normalization, coordinate correction, finger handling, and transition smoothing. Avatar identity also matters. A clip that reads as restrained on one character may read as stiff or exaggerated on another because proportions, facial rig response, and stylization change the perceptual outcome.

Latency is the final constraint. A strong directive can still arrive too late for turn-taking. The arbiter handles urgent speech-start cases with deterministic override, and the event cadence reduces redundant planning. A deployable version would benefit from caching, smaller distilled policies, local inference, and predictive triggering around silence thresholds. The evaluation must therefore report timing along with perceptual ratings.

The system also relies on accurate perception state. Voice activity detection, ASR partials, sentiment estimates, and paralinguistic summaries can all introduce errors. The directive schema gives those errors a visible trace, since the reasoner can only act on the packet it receives. Future versions should record perception confidence alongside directive confidence so social failures can be linked to upstream uncertainty when appropriate.

9 Future Work

The most immediate direction is policy distillation. The structured reasoner can log state packets, directives, traces, and rendered outcomes during normal use. Those records can train a smaller policy model that preserves the same output schema while reducing latency and removing the dependence on frontier-model API calls. The interpretability of the directive layer survives because the schema, rather than the underlying model, defines what the policy commits to.

A second direction is a direct abstraction comparison. One baseline can ask an LLM for social directives, as this thesis proposes. Another can ask for more explicit motor instructions in the style of planner-driven embodied agents such as Social Agent [16]. If both use the same retrieval and rendering stack, the study can test whether intent-level control is perceptually stronger than motor-level control for conversational avatars.

The architecture also generalizes beyond dyadic exchange. Multi-party conversation introduces gaze allocation, addressee tracking, and turn competition, while scene-aware behavior introduces documents, screens, and shared workspaces. Both extensions enter the system as additional fields in

the directive schema rather than as new modules: attention target and addressee in the multi-party case, scene reference and pointing target in the scene-aware case. The separation between social interpretation and motion realization carries over, with a richer perception state feeding a directive of the same shape.

10 Conclusion

This thesis presents LLM-reasoned motion retrieval as a control architecture for real-time conversational avatars. The system adds a structured behavioral policy between dialogue and animation. A language model reads live conversational state and emits a short-horizon directive. Retrieval and composition realize that directive through motion sources suited to speaking, listening, facial affect, and silence. A turn arbiter keeps the avatar responsive when conversational timing changes.

The primary contribution is the middle layer that gives the avatar an inspectable representation of what it is doing socially before motion is selected. The prototype demonstrates that a single typed directive can govern a live browser avatar through interruption, silence, and speech without the social policy and the motion layer needing to share representations. Prior MoMask and EMAGE work shows that the substrates the directive layer relies on are themselves runnable systems for text-to-motion generation and co-speech behavior.

A believable avatar must coordinate visible behavior across the full rhythm of an exchange, including the intervals when no one is speaking. LLM-reasoned motion retrieval offers a practical path toward that coordination by making social intent explicit before motion is selected, and by providing every layer of the system with a single ground truth reference.

References

[1] Deng, X., Gao, F., Zhang, Y., Pang, Y., Xu, X., Kang, Z., Wei, X., and Liu, Y. 2026. U-Mind: A unified framework for real-time multimodal interaction with audiovisual generation. *CVPR*.

[2] Guo, C., Mu, Y., Javed, M. G., Wang, S., and Cheng, L. 2024. MoMask: Generative masked modeling of 3D human motions. *CVPR*.

[3] Guo, C., Zou, S., Zuo, X., Wang, S., Ji, W., Li, X., and Cheng, L. 2022. Generating diverse and natural 3D human motions from text. *CVPR*.

[4] Hu, X., Yuan, H., Sang, X., Yan, B., Yu, Z., Huang, C., and Chen, K. 2026. A^2-LLM: An end-to-end conversational audio avatar large language model. *arXiv:2602.04913*.

[5] Jiang, B., Chen, X., Liu, W., Yu, J., Yu, G., and Chen, T. 2023. MotionGPT: Human motion as a foreign language. *NeurIPS*.

[6] Kalakonda, S. S., Maheshwari, S., and Sarvadevabhatla, R. K. 2024. MoRAG: Multi-fusion retrieval augmented generation for human motion. *arXiv:2409.12140*.

[7] Kopp, S., Krenn, B., Marsella, S., Marshall, A. N., Pelachaud, C., Pirker, H., and Thorisson, K. R. 2006. Towards a common framework for multimodal generation: The Behavior Markup Language. *Intelligent Virtual Agents*.

[8] Li, C., Xie, X., Cao, Y., Geiger, A., and Pons-Moll, G. 2026. FrankenMotion: Part-level human motion generation and composition. *arXiv:2601.10909*.

[9] Li, T., Bolkart, T., Black, M. J., Li, H., and Romero, J. 2017. Learning a model of facial shape and expression from 4D scans. *SIGGRAPH Asia*.

[10] Liu, H., Zhu, Z., Becherini, G., Peng, Y., Su, M., Zhou, Y., Zhe, X., Iwamoto, N., Zheng, B., and Black, M. J. 2024. EMAGE: Towards unified holistic co-speech gesture generation via expressive masked audio gesture modeling. *CVPR*.

[11] Pavlakos, G., Choutas, V., Ghorbani, N., Bolkart, T., Osman, A. A. A., Tzionas, D., and Black, M. J. 2019. Expressive body capture: 3D hands, face, and body from a single image. *CVPR*.

[12] Petrovich, M., Litany, O., Iqbal, U., Black, M. J., Varol, G., Peng, X. B., and Rempe, D. 2024. Multi-track timeline control for text-driven 3D human motion generation. *CVPR Workshop on Human Motion Generation*.

[13] Yu, Q., Tanaka, M., and Fujiwara, K. 2025. ReMoGPT: Part-level retrieval-augmented motion-language models. *AAAI*.

[14] Tevet, G., Raab, S., Gordon, B., Shafir, Y., Cohen-Or, D., and Bermano, A. H. 2023. Human Motion Diffusion Model. *ICLR*.

[15] Wang, K., Wu, Q., Song, L., Yang, Z., Wu, W., Qian, C., He, R., Qiao, Y., and Loy, C. C. 2020. MEAD: A large-scale audio-visual dataset for emotional talking-face generation. *ECCV*.

[16] Zhang, Z., Zhou, Y., Yao, H., Ao, T., Zhan, X., and Liu, L. 2025. Social Agent: Mastering dyadic nonverbal behavior generation via conversational LLM agents. *SIGGRAPH Asia*.

[17] Zhao, K., Li, G., and Tang, S. 2025. DartControl: A diffusion-based autoregressive motion model for real-time text-driven motion control. *ICLR*.